

Simple linear regression

- The sample: $\{(x_i, y_i)\}_{i=1}^n$

independent
variable

dependent
variable

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon$$

$$\varepsilon \sim N(0, \sigma^2)$$

random deviation or random error

Multiple linear regression with k independent variables. $k > 1$

- The sample: $\{(x_{1i}, x_{2i}, \dots, x_{ki}, y_i)\}_{i=1}^n$



A red arrow points from the x terms in the sample notation to the 'independent variables' box. A blue arrow points from the y term to the 'dependent variable' box.

independent
variables

dependent
variable

$$Y = \underbrace{\beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_k X_k}_{\text{True regression line}} + \varepsilon \quad \varepsilon \sim N(0, \sigma^2)$$

True regression line

random deviation or random error

Various Multiple Linear Regression Models

- **Polynomial regression model** (the independent variables are all powers of a single variable).

$$y_i = \beta_0 + \beta_1 x_i + \beta_2 x_i^2 + \cdots + \beta_p x_i^p + \varepsilon_i$$

- **Quadratic model** (polynomial regression of model of degree 2, and powers of several variables).

$$y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \beta_3 x_{1i} x_{2i} + \beta_3 x_{1i}^2 + \beta_3 x_{2i}^2 + \varepsilon_i$$

Various Multiple Linear Regression Models

- A variable that is the product of two other variables is called an **interaction**.
- These models are considered linear models, even though they contain nonlinear terms in the independent variables. The reason is that *they are linear in the coefficients, β_i* .

What is non-linear?

$$y = \beta_0 + \beta_1^2 x + \varepsilon$$

$$y = \beta_0 + e^{\beta_1 x} + \varepsilon$$

$$y = \beta_0 + \beta_1 x^{\beta_2} + \varepsilon$$

Estimating the Coefficients

- In any multiple regression model, the estimates

$$\hat{\beta}_0, \hat{\beta}_1, \dots, \hat{\beta}_p$$

are computed by least-squares, just as in simple linear regression. The equation

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \dots + \hat{\beta}_p x_p$$

is called the **least-squares equation** or **fitted regression equation**.

- Now define $\hat{y}_i$ to be the y coordinate of the least-squares equation corresponding to the x values $(x_{1i}, \dots, x_{pi})$.

Estimating the Coefficients

- The residuals are the quantities.

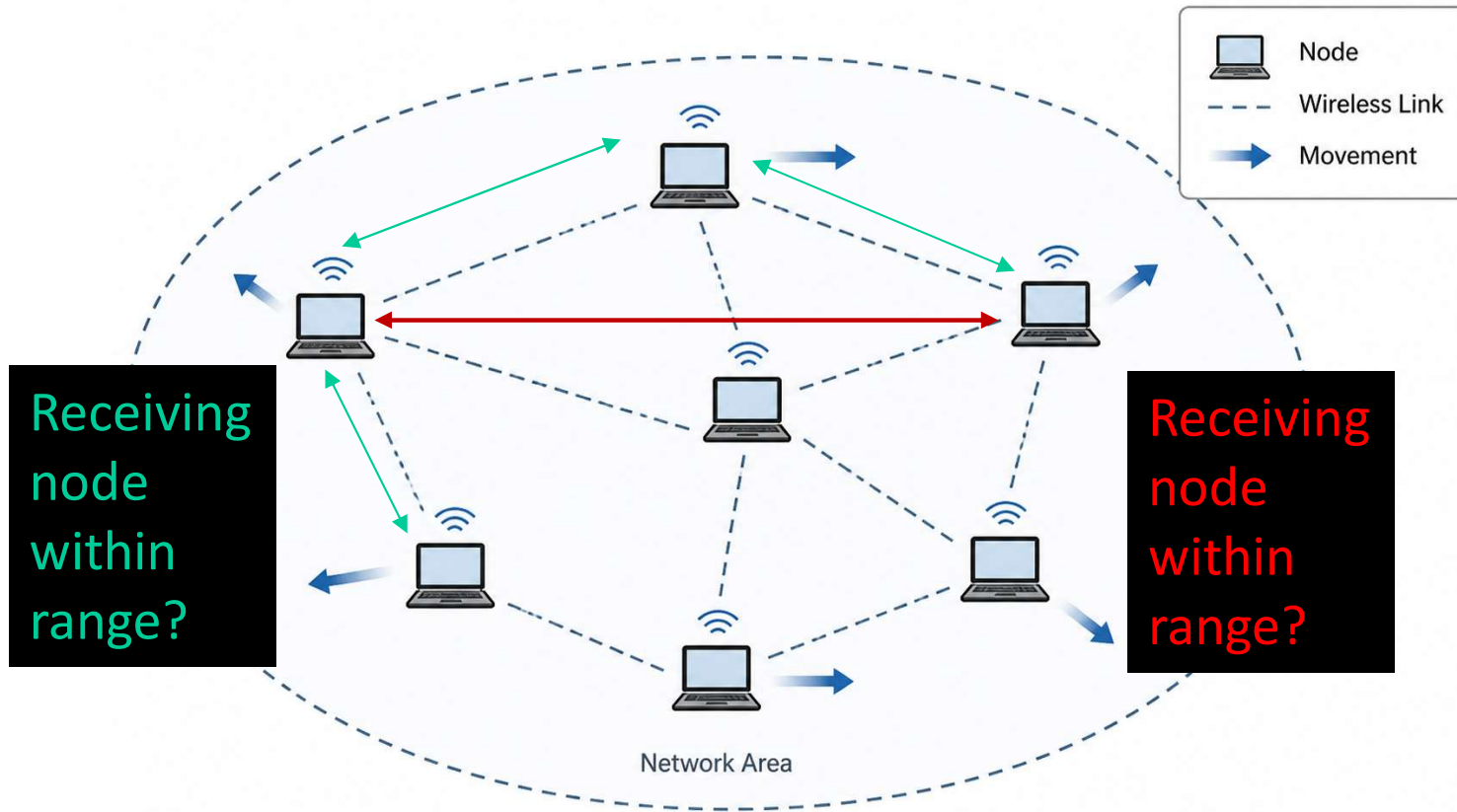
$$e_i = y_i - \hat{y}_i$$

which are the differences between the observed y values and the y values given by the equation.

- We want to compute $\hat{\beta}_0, \hat{\beta}_1, \dots, \hat{\beta}_p$ so as to minimize the sum of the squared residuals. The formulas are complicated, so we rely on computers to calculate them.

Example:

A mobile ad hoc computer network



Goodput: the proportion of messages that will be successfully delivered

Can we predict the Goodput?

Potential factors:

- the average node speed
- the length of time that the nodes pause at each destination.

Example:

A mobile ad hoc computer network

Data was collected on average node speed, average pause time, and goodput for 25 simulated mobile ad hoc networks.

Consider the following model:

$$\text{Goodput} = \beta_0 + \beta_1 \text{Speed} + \beta_2 \text{Pause} + \beta_3 \text{Speed} \cdot \text{Pause} \\ + \beta_4 \text{Speed}^2 + \beta_5 \text{Pause}^2 + \varepsilon$$

Computer Output Part 1

$$\text{Goodput} = 96.0 - 1.82 \text{ Speed} + 0.565 \text{ Pause} + 0.0247 \text{ Speed} * \text{Pause} + 0.0140 \text{ Speed}^2 - 0.0118 \text{ Pause}^2$$

Predictor	Coef	SE Coef	T	P
Constant	96.024	3.946	24.34	0.000
Speed	-1.8245	0.2376	-7.68	0.000
Pause	0.5652	0.2256	2.51	0.022
Speed*Pa	0.024731	0.003249	7.61	0.000
Speed^2	0.014020	0.004745	2.95	0.008
Pause^2	-0.011793	0.003516	-3.35	0.003

$S = 2.942$ $R - Sq = 93.2\%$ $R - Sq(\text{adj}) = 91.4\%$

Note: sample size is 25 (n=25), number of estimated parameters is 6 (p=6).

Individual Significance Test

Hypothesis testing on β_i

$$H_0: \beta_i = 0 \quad \text{Test statistics: } T = \frac{b_i}{S_{b_i}} \sim t_{n-(k+1)}$$

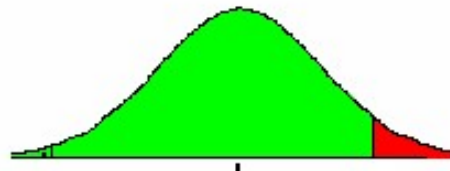
$$H_a: \beta_i > 0$$

$$H_a: \beta_i \neq 0$$

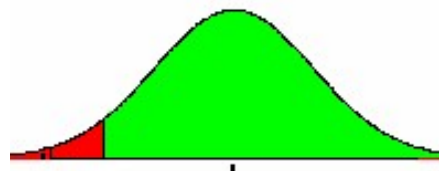
$$H_a: \beta_i < 0$$

Accept

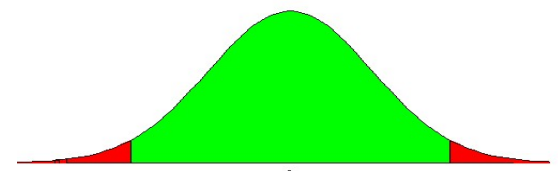
Reject



$$\text{R.R.: } t \geq t_\alpha$$



$$\text{R.R.: } t \leq -t_\alpha$$



$$\begin{aligned} \text{R.R.: } t &\leq -t_{\alpha/2} \\ \text{or } t &\geq t_{\alpha/2} \end{aligned}$$

Computer Output Part 1

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$$H_0: \beta_k = 0$$

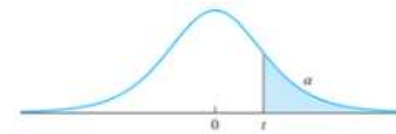
$$H_a: \beta_k \neq 0$$

$$T = \frac{b_k}{s_{b_k}} \sim t_{n-k-1}$$

S = 2.942 R – Sq = 93.2% R – Sq(adj) = 91.4%

Note: sample size is 25 (n=25), number of independent variables is 5 (k=5).

Upper percentage points for the Student's t distribution



ν	α								
	0.40	0.25	0.10	0.05	0.025	0.01	0.005	0.001	0.0005
1	0.325	1.000	3.078	6.314	12.706	31.821	63.657	318.309	636.619
2	0.289	0.816	1.886	2.920	4.303	6.965	9.925	22.327	31.599
3	0.277	0.765	1.638	2.353	3.182	4.541	5.841	10.215	12.924
4	0.271	0.741	1.533	2.132	2.776	3.747	4.604	7.173	8.610
5	0.267	0.727	1.476	2.015	2.571	3.365	4.032	5.893	6.869
6	0.265	0.718	1.440	1.943	2.447	3.143	3.707	5.208	5.959
7	0.263	0.711	1.415	1.895	2.365	2.998	3.499	4.785	5.408
8	0.262	0.706	1.397	1.860	2.306	2.896	3.355	4.501	5.041
9	0.261	0.703	1.383	1.833	2.262	2.821	3.250	4.297	4.781
10	0.260	0.700	1.372	1.812	2.228	2.764	3.169	4.144	4.587
11	0.260	0.697	1.363	1.796	2.201	2.718	3.106	4.025	4.437
12	0.259	0.695	1.356	1.782	2.179	2.681	3.055	3.930	4.318
13	0.259	0.694	1.350	1.771	2.160	2.650	3.012	3.852	4.221
14	0.258	0.692	1.345	1.761	2.145	2.624	2.977	3.787	4.140
15	0.258	0.691	1.341	1.753	2.131	2.602	2.947	3.733	4.073
16	0.258	0.690	1.337	1.746	2.120	2.583	2.921	3.686	4.015
17	0.257	0.689	1.333	1.740	2.110	2.567	2.898	3.646	3.965
18	0.257	0.688	1.330	1.734	2.101	2.552	2.878	3.610	3.922
19	0.257	0.688	1.328	1.729	2.093	2.539	2.861	3.579	3.883
20	0.257	0.687	1.325	1.725	2.086	2.528	2.845	3.552	3.850
21	0.257	0.686	1.323	1.721	2.080	2.518	2.831	3.527	3.819
22	0.256	0.686	1.321	1.717	2.074	2.508	2.819	3.505	3.792
23	0.256	0.685	1.319	1.714	2.069	2.500	2.807	3.485	3.768
24	0.256	0.685	1.318	1.711	2.064	2.492	2.797	3.467	3.745
25	0.256	0.684	1.316	1.708	2.060	2.485	2.787	3.450	3.725

T	P
24.34	0.000
-7.68	0.000
2.51	0.022
7.61	0.000
2.95	0.008
-3.35	0.003

Interpreting Output

Much of the output is analogous to that of simple linear regression.

1. The fitted regression equation is presented near the top of the output.
2. Below that the coefficient estimates and their estimated standard deviations.
3. Next to each standard deviation is the Student's t statistic for testing the null hypotheses that the true value of the coefficient is equal to 0.
4. The P -values for the tests are given in the next column.

Multiple regression with k independent variables. $k > 1$:

<i>Source</i>	<i>df</i>	<i>Sum Sq.</i>	<i>Mean Sum Sq.</i>	<i>F</i>
<i>Regression</i>	k	SSR	$MSR = \frac{SSR}{k}$	$\frac{MSR}{MSE}$
<i>Error</i>	$n-(k+1)$	SSE	$MSE = S^2 = \frac{SSE}{n - (k + 1)}$	
<i>Total</i>	$n-1$	SST		

Sums and Mean Sums

Let all $\beta_i=0$.

k: number of
independent
variables

$$MSE = S^2 = \frac{SSE}{n - (k + 1)} \sim \chi_{n-(k+1)}^2$$

$$MSR = \frac{SSR}{k} \sim \chi_k^2$$

What is the distribution of $\frac{SSR/k}{SSE/n-(k+1)}$?

$$\frac{W/u}{Y/v} \sim F_{u,v} \quad W \sim \chi_u^2, \quad Y \sim \chi_v^2$$

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<i>Error</i>	$n-(k+1)$	SSE	$MSE = S^2 = \frac{SSE}{n - (k + 1)}$	
<i>Total</i>	$n-1$	SST		

$H_0: \beta_i = 0$ for all i
 $H_a: \beta_i \neq 0$ for at least one i

$\xrightarrow{\text{can be tested via:}}$

reject H_0 if $F > F_{\alpha, k, n-(k+1)}$.

Overall significance test

Computer Output Part 2

Analysis of Variance

Source	DF	SS	MS	F	P
Regression	5	2240.49	448.10	51.77	0.000
 Residual Error	19	164.46	8.66		
Total	24	2404.95			

$$H_0: \beta_i = 0 \text{ for all } i$$

$$H_a: \beta_i \neq 0 \text{ for at least one } i$$

Note: sample size is 25 (n=25), number of independent variables is 5 (k=5).

ν_2	α	ν_1								
		1	2	3	4	5	6	7	8	9
10	0.100	3.29	2.92	2.73	2.61	2.52	2.46	2.41	2.38	2.35
	0.050	4.96	4.10	3.71	3.48	3.33	3.22	3.14	3.07	3.02
	0.010	10.04	7.56	6.55	5.99	5.64	5.39	5.20	5.06	4.94
	0.001	21.04	14.91	12.55	11.28	10.48	9.93	9.52	9.20	8.96
18	0.100	3.01	2.62	2.42	2.29	2.20	2.13	2.08	2.04	2.00
	0.050	4.41	3.55	3.16	2.93	2.77	2.66	2.58	2.51	2.46
	0.010	8.29	6.01	5.09	4.58	4.25	4.01	3.84	3.71	3.60
	0.001	15.38	10.39	8.49	7.46	6.81	6.35	6.02	5.76	5.56
19	0.100	2.99	2.61	2.40	2.27	2.18	2.11	2.06	2.02	1.98
	0.050	4.38	3.52	3.13	2.90	2.74	2.63	2.54	2.48	2.42
	0.010	8.18	5.93	5.01	4.50	4.17	3.94	3.77	3.63	3.52
	0.001	15.08	10.16	8.28	7.27	6.62	6.18	5.85	5.59	5.39

F

P

51.77 0.000

If $H_0: \beta_i = 0$ for all i ,

$$\frac{SSR/k}{SSE/n - (k + 1)} \sim F_{k, n-k-1}$$

Analysis of Variance Table

5. The D F column gives the degrees of freedom, the degrees of freedom for regression is equal to the number of independent variables in the model. The degrees of freedom for “Residual Error” is the number of observations – number of parameters estimated. The total degrees of freedom is the sum of the degrees of freedom for regression and for error.
6. The next column is S S. This column gives the sum of squares, the first is regression sum of squares, $S S R$, the second is error sum of squares, $S S E$, and the third is the total sum of squares, $S S T = S S R + S S E$.

More on the ANOVA Table

7. The column MS is the column with the mean sum of squares which is the sums of squares divided by their respective degrees of freedom. Note that the mean square error is equal to the variance estimate, S^2 .
8. The column labeled F presents the mean square for regression divide by the mean square for error.
9. This is the F statistic that we discussed earlier that is used for testing the null hypothesis that none of the independent variables are related to the dependent variable.

Interpretation of r^2 in multiple regression with k independent variables

$$r^2 = 1 - \frac{SSE}{SST} = \frac{SSR}{SST} \quad \text{coefficient of multiple determination}$$

As you add more variables: $r^2 \uparrow$

But we lose one df for each additional parameter: not necessary a better model.

We need to check on MSE as well: $MSE = \frac{SSE}{n-k-1} \downarrow?$

We can add new independent variables to explain Y . For example if we have n observations and $n-1$ variables:

$$Y_j = \beta_0 + \beta_1 X_{1j} + \beta_2 X_{2j} + \dots + \beta_{n-1} X_{n-1j} \quad j = 1, \dots, n$$

$\Rightarrow$ an almost perfect fit as $r^2 = 1$, but $MSE \uparrow \infty$.

Evaluation criteria for “best”

r^2 is adjusted for the number of parameters in the model:

$$r_a^2 = 1 - \frac{SSE/(n - k)}{SST/(n - 1)} \quad \text{adjusted coefficient of multiple determination}$$

Equivalent to minimizing $S^2 = \text{MSE}$.

r_a^2 is also interpreted as the percentage of variability explained in the dependent variable Y.

Using the Output

- Use the fitted regression equation to predict for future observations.
- Calculate residuals for a value of y .
- Construct a confidence interval for the coefficient of the independent variables.

Example: Water Consumption

A production plant cost-control engineer is responsible for cost reduction. One of the costly items in his plant is the amount of water used by the production facilities each month. He decided to investigate water usage by collecting 17 observations on his plant's water usage and other variables. All his data as well as MINITAB outputs corresponding to several regression models are given below.

Definition of variables

X_1 = daily amount of
production (in pounds)

X_2 = average daily
temperature (°F)

X_3 = number of working
machines on a day

X_4 = number of employees
working on a day

Y = the daily water usage
(gallons)

	X_1	X_2	X_3	X_4	Y
1	7107	58.8	21	129	3067
2	6373	65.2	22	141	2828
3	6796	70.9	22	153	2891
4	9208	77.4	20	166	2994
5	14792	79.3	25	193	3082
6	14564	81	23	189	3898
7	11964	71.9	20	175	3502
8	13526	63.9	23	186	3060
9	12656	54.5	20	190	3211
10	14119	39.5	20	187	3286
11	16691	44.5	22	195	3542
12	14571	43.6	19	206	3125
13	13619	56	22	198	3022
14	14575	64.7	22	192	2922
15	14556	73	21	191	3950
16	18573	78.9	21	200	4488
17	15618	79.4	22	200	3295

Regression Analysis:

y versus x1; x2; x3; x4

$$y = 6360 + 0.212 x_1 + 13.9 x_2 - 127 x_3 - 21.8 x_4$$

Predictor	Coef	SE Coef	T	P
Constant	6360	1314	4.84	0.000
x1	0.2117	0.04554	4.65	0.001
x2	13.869	5.160	2.69	0.020
x3	-126.69	48.02	-2.64	0.022
x4	-21.818	7.285	-3.00	0.011
S = 249.0		R-Sq = 76.7%	R-Sq(adj) = 68.9%	

Analysis of Variance

Source	DF	SS	MS	F	P
Regression	4	2448834	612209	9.88	0.001
Residual Error	12	743798	61983		
Total	16	3192632			

	X1	X2	X3	X4
Min	6373	39.5	19	129
Max	18573	81	25	206
Average	12900	64.85	21.47	182
St. Dev	3527	13.51	1.46	22

X_1 = daily amount of production (in pounds)
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Questions

- Is this a significant regression model?

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- How much of the variability in daily water usage is explained by this model? Is this sufficient?

Regression Analysis:

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- Is this a significant regression model?
- Explain the meaning of the regression model: how does each independent variable affect daily water usage? Do they increase or decrease the daily water usage? Are their effects significant?
- How much of the variability in daily water usage is explained by this model? Is this sufficient?
- The management aims to increase the production to a level of 20000 pounds, which may require the use of 25 machines and 250 employees. Can this model be used to predict the daily water usage in this case? Why or why not? Explain.

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Max	18573	81	25	206
Average	12900	64.85	21.47	182
St. Dev	3527	13.51	1.46	22

Example: A Final Exam Question

- A worldwide organization wants to estimate the literacy ratio, LR, in any given settlement district, where LR is:

$$LR = \frac{\textit{number of people who can read \& write}}{\textit{total population size}}$$

- Their first step of analysis uses an economic indicator, EI, to generate rough predictions of literacy ratio. The economic indicator can take continuous values in scale (0,10), where 0 is the worst possible economic situation and 10 is the best.

Example:

$$r^2 = 1 - \frac{SSE}{SST} = \frac{SSR}{SST}$$

A Final Exam Question

Regression Analysis: LR versus EI

More variables?

The regression equation is

LR = 0.622 + 0.0392 EI

Urban vs Rural?

Predictor	Coef	SE Coef	T	P
Constant	0.62206	0.02445	25.44	0.000
EI	0.039221	0.007856	4.99	0.000

S = 0.0885719 R-Sq = 24.2% R-Sq(adj) = 23.2%

Analysis of Variance

Source	DF	SS	MS	F	P
Regression	1	0.19551	0.19551	24.92	0.000
Residual Error	78	0.61191	0.00784		
Total	79	0.80742			

Categorical or dummy variables

- Environment
- Man/Woman/Children
- ...

Example: A Final Exam Question

- After analyzing the above regression model, and considering the rest of data they have, the statistical team of the organization suggested another regression model, which now indicates whether the settlement is an urban (city) or rural (country) area.

$$U - R = \begin{cases} 1 & \text{if urban} \\ 0 & \text{if rural} \end{cases}$$

Example: A Final Exam Question

Regression Analysis: LR versus EI, U-R

The regression equation is

$$\text{LR} = 0.576 + 0.0356 \text{ EI} + 0.0961 \text{ U-R}$$

Predictor	Coef	SE Coef	T	P
Constant	0.57575	0.02232	25.80	0.000
EI	0.03565	0.00669	5.33	0.000
U-R	0.09613	0.01714	5.61	0.000

S = 0.0751069 R-Sq = 46.2% R-Sq(adj) = 44.8%

Analysis of Variance

Source	DF	SS	MS	F	P
Regression	2	0.37306	0.18653	33.07	0.000
Residual Error	77	0.43436	0.00564		
Total	79	0.80742			

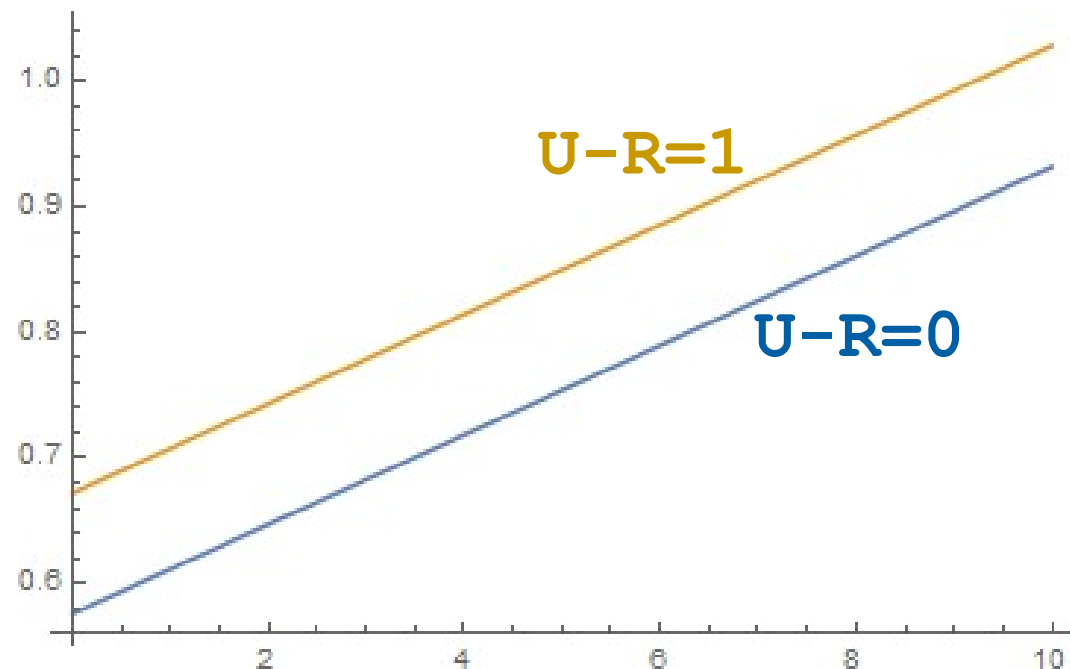
Example: A Final Exam Question

Regression Analysis: LR versus EI, U-R

The regression equation is

$$LR = 0.576 + 0.0356 EI + 0.0961 U-R$$

$$U - R = \begin{cases} 1 & \text{if urban} \\ 0 & \text{if rural} \end{cases}$$

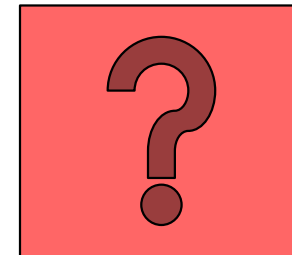


What does this equation mean?

Example: A Final Exam Question

- What if we wanted to check the effects of continents (Europe, Asia, North America) on literacy ratio, LR?

$$C = \begin{cases} 0 & \text{Europe} \\ 1 & \text{Asia} \\ 2 & \text{North America} \end{cases}$$



$$LR = \beta_0 + \beta_{EI} EI + \beta_{U-R} U-R + \beta_C C + \varepsilon$$

Is the effect of being in Europe instead of Asia equal to the effect of being in Asia instead of North America?

Example: A Final Exam Question

- What if we wanted to check the effects of continents (Europe, Asia, North America) on literacy ratio, LR?

$$E = \begin{cases} 1 & \text{if in Europe} \\ 0 & \text{Otherwise} \end{cases}$$

~~$$N = \begin{cases} 1 & \text{if in North America} \\ 0 & \text{Otherwise} \end{cases}$$~~

$$A = \begin{cases} 1 & \text{if in Asia} \\ 0 & \text{Otherwise} \end{cases}$$

$$LR = \beta_0 + \beta_{EI} EI + \beta_{U-R} U-R + \beta_E E + \beta_A A + \cancel{\beta_N N} + \varepsilon$$

Do we need all three?

Example: A Final Exam Question

- What if we wanted to check the effects of continents (Europe, Asia, North America) on literacy ratio, LR?

$$E = \begin{cases} 1 & \text{if in Europe} \\ 0 & \text{Otherwise} \end{cases} \quad A = \begin{cases} 1 & \text{if in Asia} \\ 0 & \text{Otherwise} \end{cases}$$

$$LR = \beta_0 + \beta_{EI} EI + \beta_{U-R} U-R + \beta_E E + \beta_A A + \varepsilon$$

Europe	E=1, A=0
Asia	E=0, A=1
North America	E=0, A=0

- Meanings of β_E and β_A ?
- What is the effect of being in Europe instead of Asia?